

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Abstract

LLM-generated medication alerts can sound more certain than their evidence permits. We built a deterministic pre-display monitor for one workflow: pharmacogenomic and genomic medication-alert text. The monitor checks source support, population fit, and claim strength, then returns allow, narrow, abstain, or deny. Stage 1 is a synthetic specification test: the case bank supplies the structured annotations that a future live system would need to extract, and ablation tests whether any check is dispensable. Disabling source support let 2 of 23 designed overclaims pass unchanged; disabling population fit let 1 pass; disabling claim strength let 6 pass. With all checks active, all 23 designed overclaim archetypes were blocked or narrowed, and all 10 bounded alerts passed unchanged. The first-triggering check matched the author-assigned check label in 31 of 33 cases. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Keywords: clinical decision support; large language models; pharmacogenomics; evidence overclaiming; model evaluation

Data and Code Availability The Stage 1 case bank is synthetic and contains no real patient records, protected health information, or private genomic data. For anonymous venue review, code and synthetic data should be supplied as an anonymized supplemental archive; the de-anonymized public repository may be disclosed after acceptance or in course-facing materials.

Institutional Review Board (IRB) Stage 1 uses only author-designed synthetic cases and public guideline/database names, and therefore does not require IRB approval. Any Stage 2 use

of clinical cases or expert adjudication would require appropriate local governance.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a claim for a different population, or a surrogate endpoint into patient-outcome benefit. Here, transportability means applying evidence from one source population as if it supported the same claim in a different target population. The gap matters because fluency can hide what evidence does, and does not, authorize. Pharmacogenomic medication alerts are a useful testbed because CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation ([Clinical Pharmacogenetics Implementation Consortium, 2026](#); [ClinPGx, 2026a](#)). The thesis tested here is deliberately narrow and disputable: overclaim control in this workflow needs three separable checks—source support, population fit, and claim strength—rather than a single endpoint-evidence rule.

2. Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne et al. emphasize that outcome claims require pragmatic clinical evaluation ([Byrne et al., 2024](#)), while Stegenga motivates caution when evidentiary pipelines make interventions appear stronger than they are ([Stegenga, 2018](#)). FDA–NIH BEST materials provide endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates,

and clinical outcomes are not interchangeable (U.S. Food and Drug Administration and National Institutes of Health, 2016). Pharmacogenomics adds a positive control: CPIC-style resources can support bounded prescribing language, including CYP2C19–clopidogrel and DPYD–fluoropyrimidine examples (Lee et al., 2022; Amstutz et al., 2018), but do not automatically support claims that an LLM improves outcomes, that an association is causal, or that a result generalizes across populations without recalibration.

3. Related Work

LLM factuality methods such as SelfCheckGPT test whether generated content is consistent across black-box samples (Manakul et al., 2023). Selective classification supplies the decision vocabulary for abstention: a system may decline to answer when coverage is less important than risk control (Geifman and El-Yaniv, 2017). Clinical alert-fatigue work explains why this choice matters in practice: repeated and low-information alerts can reduce clinician response (Ancker et al., 2017). This study tests a smaller executable claim. Given structured annotations about source support, target population, and claim strength, a pre-display monitor can separate designed pharmacogenomic overclaims from bounded alerts while showing which checks an endpoint-only rule would miss.

4. Methods

The controller is a deterministic pre-display monitor, not an autonomous clinical agent. A Stage 1 case contains a candidate alert, source-status annotation, population-fit annotation, endpoint annotation, actionability annotation, and expected review label. The monitor reads those annotations in a fixed order: source support, population fit, then claim strength. It returns one of five actions: allow, narrow, abstain (population fit), abstain (source support), or deny. The repository preserves the implementation identifiers for the three checks, but the paper uses their clinical names.

5. Stage 1 Scaffold

The case bank contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty, obsolete evidence, and changed clinical context. An isolating case is designed so that only one check changes the action; with that check disabled, the overclaim passes unchanged. Within the 23 overclaim archetypes, three cases isolate a fabricated CPIC addendum attached to an otherwise actionable claim, a real guideline-supported claim applied to a mismatched population, and a stale but real guideline used as current evidence. The 23/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them. The three checks are summarized in Table 1. The decisive test is ablation, not action conformance: if a disabled check lets an overclaim pass unchanged, that check is doing work not captured by the remaining checks.

With all checks active, all 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged. Table 2 shows that disabling each check allows at least one designed overclaim to pass unchanged, so source support and population fit are not reducible to claim strength. Removing source support allows 2/23 overclaims through unchanged; removing population fit allows 1/23 through; removing claim strength allows 6/23 through. No check is redundant in this synthetic case bank: each check blocks at least one designed overclaim that passes unchanged when that check is disabled.

Primary-check conformance, defined as the first enabled check matching the author-assigned check label, was 31/33. In PGX19 and PGX24, the controller returned the expected action, but the triggering check differed from the author-assigned label because source support has precedence. These two disagreements are check-attribution errors.

6. Evaluation Plan

Stage 2 turns the specification test into a held-out reader study. Independent case authors write

Table 1: Stage 1 decision predicates implemented by the controller.

Check	Predicate used in Stage 1	Monitor action
Source support	The cited source cannot be verified.	Deny the alert for unverifiable source support.
Source support	The source is real but conflicting, uncertain, outdated, or applied after the clinical context has changed.	Abstain for source support.
Source support	Source support is absent. If the medication action is also unsupported, deny; otherwise abstain.	Deny unsupported action or abstain.
Population fit	Target-population fit is mismatched or cannot be determined after source support passes.	Abstain (population fit).
Claim strength	The requested medication action is not supported by the evidence.	Deny unsupported action.
Claim strength	The evidence supports only a context-dependent, associative, mechanistic, algorithmic, surrogate, or uncertain-variant claim.	Narrow the alert.
None	Earlier predicates do not fire.	Allow bounded alert.

Table 2: One-check ablation on the synthetic case bank.

Disabled check	Allowed	Action Δ	Check Δ
Source support	2	7	12
Population fit	1	5	5
Claim strength	6	6	6

pharmacogenomic alert scenarios without seeing the decision table. The ungated arm shows the LLM draft unchanged; the gated arm shows the same draft after structured annotations pass through the monitor. Two clinical reviewers assign binary labels for overclaim present, denial appropriate, clinician authority preserved, population fit adequate, and source support adequate; they also select one error class from a fixed taxonomy. The primary endpoint is the difference in reviewer-labeled overclaim rate between arms. The safety endpoint is the difference in inappropriate-denial rate; the monitor is useful only if overclaiming falls without a compensating rise in inappropriate denial.

7. Ethics and Limitations

Transparency The monitor produces a decision trace. For each alert, the trace names the structured evidence feature that triggered allow, narrow, abstain, or deny. Clinicians and review-

ers can inspect the rule that changed, blocked, or preserved the displayed alert.

Fairness Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language depends on ancestry and tested population, founder-variant evidence may not generalize to the US population, and biobank results may be unrepresentative for underrepresented or recently admixed groups (ClinPGx, 2026b; Genome India Project, 2026). Cases PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint. PGX24 represents the high-risk practice of using a polygenic score to deny opioid or stimulant therapy.

Governance and limitations Stage 1 contains no PHI and makes no diagnosis or treatment recommendation; deployed clinical decision support would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging. The key limitation is that results are synthetic and author-designed. They show rule conformance, not generalization, and assume structured annotations rather than extracting them from live LLM output or EHR context.

8. Conclusion

The practical problem is annotation, not compute. Stage 1 shows that the monitor logic is auditable and that the three checks do different work on a synthetic pharmacogenomic case

bank. Source support, population fit, and claim strength each block at least one designed over-claim that passes when that check is removed. That result turns the next experiment into a concrete measurement problem: determine whether live LLM and EHR pipelines can populate the three annotations reliably enough to reduce over-claiming without increasing inappropriate denial.

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